JMEPEG (2013) 22:1680–1687
DOI: 10.1007/s11665-012-0465-z

1059-9495/$19.00

Thermal Shock Properties of a 2D-C/SiC Composite Prepared by Chemical Vapor Infiltration

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(*Submitted April 23, 2012; in revised form December 7, 2012; published online December 22, 2012*)

The thermal shock properties of a two-dimensional carbon fiber-reinforced silicon carbide composite with a multilayered self-healing coating (2D-C/SiC) were investigated in air. The composite was prepared by low-pressure chemical vapor infiltration. 2D-C/SiC specimens were thermally shocked for different cycles between 900 and 300 °C. The thermal shock resistance was characterized by residual tensile properties and mass variation. The change of the surface morphology and microstructural evolution of the composite were examined by a scanning electron microscope. In addition, the phase evolution on the surfaces was identified using an X-ray diffractometer. It is found that the composite retains its tensile strength within 20 thermal shock cycles. However, the modulus of 2D-C/SiC decreases gradually with increasing thermal shock cycles. Extensive pullout of fibers on the fractured surface and peeling off of the coating suggest that the damage caused by the thermal shock involves weakening of the bonding strength of coating/composite and fiber/matrix. In addition, the carbon fibers in the near-surface zone were oxidized through the matrix cracks, and the fiber/matrix interfaces delaminated when the composite was subjected to a larger number of thermal shock cycles.

Keywords C/SiC composite, oxidation, residual strength, thermal shock

1. Introduction

Carbon fiber-reinforced silicon carbide composites (C/SiC) do not suffer from the brittleness and low reliability associated with monolithic ceramics. Therefore, they are widely used as structural materials in different fields including advanced engines, gas turbines for power/steam co-generation, heat exchangers, as well as in the development of future nuclear reactors (Ref 1-4). However, the carbon fibers or the carbon interface between matrix and fiber in the composites can be easily oxidized if the composites are exposed to high temperature in oxidizing environments. The oxidation leads to degradation of the mechanical properties and shortens the lifetime of the composite. The situation becomes more severe when the components, made of C/SiC, are used in thermomechanical environments where the thermal shock/transient thermal conditions (a sudden change in temperature) are often inevitable. Compared with monolithic ceramics, continuous fiber-reinforced ceramic composites display much more complex behaviors under thermal transient conditions (Ref 5). For example, the fiber reinforcement can change the crack initiation sites and propagation paths in the composites under thermal shock. Meanwhile, additional thermal stresses can be introduced because of the thermal expansion mismatch between fibers and matrix. Moreover, many factors, such as the type of fiber/matrix, fiber architecture, interfacial characteristics, and matrix-forming processes, also affect their thermal shock behaviors (Ref 6-10). Therefore, the thermal shock resistance of ceramic composites reinforced with continuous fibers cannot be explicitly evaluated in the simple ways that are used for monolithic ceramics (Ref 11). It is postulated that the thermal shock resistance of fiber-reinforced brittle matrix composites could be given by the critical temperature difference ($\Delta T_{\max}$) that caused matrix cracking (Ref 12, 13). Kastritseas et al. (Ref 14) developed a semi-empirical formula of $\Delta T_{\max}$ that initiates fracture for unidirectional fiber-reinforced ceramic composites. The formula took the anisotropic stress field generated during the thermal shock, the processing temperature, the quenching temperature, and material properties into consideration. Owing to the difficulty in calculating the thermal stresses accurately in the composites, the reported values varied over a wide range. Therefore, the abilities to resist thermal shock were always evaluated by measuring the residual strength or modulus, or observing the damage within the composites. It was found by Yin et al. (Ref 15) that the thermal shock generated several matrix cracks in a 3D-C/SiC. The flexural strength degraded gradually until the cracks were saturated in the matrix. Mei et al. (Ref 16, 17) found that the thermal strain of C/SiC was closely related to the cyclic temperature. Furthermore, the cyclic stresses could lead to cracking in coating and matrix, debonding of fiber/matrix interface, and interface sliding. In addition, the oxygen in the atmosphere could aggravate the damage by alternate oxidation between internal and external fibers in the composites. Fujii and Yamada (Ref 18) indicated that the reduced mismatch

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between the carbon and SiC could improve the thermal shock resistance of a SiC compositionally graded C/C composites. The materials in above thermal shock processes were cooled either by an environment (air or water) or a controlled atmosphere. Actually, the components, made of C/SiC, always connected with metal parts. The conduction of heat in C/SiC through the metal parts played a more important role in the temperature change than the heat exchange by convection or radiation.

The thermal shock behaviors of a two-dimensional (2D) carbon fiber-reinforced silicon carbide composite with a multilayered self-healing coating (2D-C/SiC) were investigated between 300 and 900 °C in air in the present study. The specimens were cooled through an iron plate. The thermal shock resistance of 2D-C/SiC was characterized by residual tensile properties and mass variation. Also, the microstructures of the thermally shocked specimens were examined to reveal the damage caused by the thermal shock.

2. Experimental Procedures

2.1 Materials

In 2D-C/SiC, stacking plain weave carbon fiber cloths were used for the reinforcement (1 K, T-300, Toray Co, Japan). A low-pressure chemical vapor infiltration (CVI) process was used to prepare the composites. The pyrolytic carbon (P_yC) interphase with a thickness of 0.2 μm was deposited on carbon fibers with C_3H_6 at 1173 K and 5 kPa. Methyltrichlorosilane (MTS, CH_3SiCl_3) was used for depositing the SiC matrix. The applied temperature, the total pressure, and the molar ratio of H_2 to MTS were 1000 °C, 5 kPa, and 10, respectively. After the deposition of SiC, the contoured, edge-loaded test specimens, shown in Fig. 1, were machined from the fabricated plates along one of the weaving directions. In order to improve the oxidation resistance of the composite, the specimens were further coated with a multilayered coating (B_4C/SiC/B_4C/SiC) of about 50-μm thickness, shown in Fig. 2. B_4C in SiC coating could improve the oxidation resistance at high temperatures (Ref 4). B_4C and SiC in the multilayered coating was produced by alternate CVI. The preparation process of SiC was same as that used for the SiC matrix. B_4C was deposited using BCl_3, C_3H_6, and H_2 mixtures with a molar ratio of 1:1:5. The applied deposition temperature and pressure were 1000 °C and 3 kPa, respectively. The volume fraction of the fibers and density of the 2D-C/SiC were 40% and 2.05 g/cm^3, respectively.

2.2 Procedures for the Thermal Shocks

A tube furnace was employed to carry out the thermal shock experiments. It consists of a cylindrical alumina tube surrounded by heating coils, which are made of Nichrome 80/20 and embedded in a thermally insulating matrix. Temperature was controlled via feedback from a Type K thermocouple. The thermal shock experiments were conducted by alternating the specimens quickly between 900 and 300 °C in static air. Each specimen was held in the tube furnace, which was preheated to 900 °C, for 10 minutes and then cooled to 300 °C within 15 s by means of an iron plate. The estimated cooling rate was about 40 °C/s. After the temperature of the specimen dropped to 300 °C, it was immediately sent back to the furnace, temperature of which was held at 900 °C, for the next heating-cooling cycle. The time to move the specimen from the furnace to the iron plate was approximately 2 s. Such a heating-cooling cycle indicated that the specimens were thermally shocked for one time. In order to reveal the thermal shock resistance of the composite, at least three specimens were investigated at prescribed number of cycles (20, 40, and 60).

2.3 Measurements of the Residual Tensile Properties and Mass Variation

The tensile properties were measured using a mechanical testing machine (Model: CSS-280S-100, Changchun, China) at RT. The procedure for measuring the residual tensile properties was in accordance with ASTM 1275 (Ref 19). The tensile experiments were conducted when the specimens, shown in Fig. 1, were pulled into the wedge of the grip. To guarantee the specimen alignment, lateral centering of the test specimen within the grip attachments was accomplished by wedge-type inserts machined to fit within the grip cavity. The applied displacement rate was 0.5 mm/min. The strain was recorded by an extensometer (Epsilon 3548, USA) with a gauge length of 25 mm. The tensile strength and modulus of the unshocked material were determined for five specimens, and the properties of the specimens thermally shocked at prescribed number of cycles (20, 40, and 60) were measured for three specimens. The normalized residual strength (b) was obtained by the equation:

$$b = \frac{R_1}{R_0}$$

where R_1 is the average tensile strength of the shocked specimen, and R_0 is the average tensile strength of the unshocked

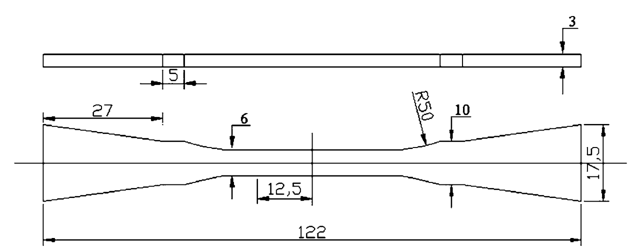

Fig. 1 Dimensions of the specimens used for the thermal shock experiments (in mm)

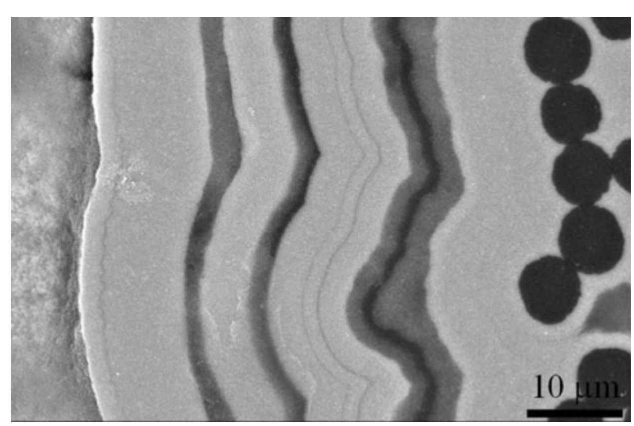

Fig. 2 Microstructure of the multilayered coating, composed of SiC (gray) and BxC (dark)

specimens. The residual modulus was determined by the slope of the initial linear portion of the tensile stress-strain curves.

The mass of the cyclically shocked specimens was measured by means of a balance (AG204, Mettler Toledo, Switzerland) with a precision of 0.1 mg. The mass variation (a) was calculated by the following equation:

$$a = \frac{m_1 - m_0}{m_0} \times 100\%$$

where m_1 stands for the mass of the specimens thermally shocked for 0, 1, 6, 12, 20, 40, or 60 cycles; and m_0 represents the mass of the unshocked specimen.

2.4 Microstructural Examination

Fractured morphologies and microstructure were examined by a scanning electron microscope (SEM, S2700, Hitachi, Japan) operated at 20 kV in backscattered electron mode. The specimens used for examining microstructure of the thermally shocked specimens were cut from the gauge section. The composition was determined by means of an energy dispersive X-ray spectroscope (EDS, Oxford Instruments, ISIS300, England) attached in the SEM. The phases on the surface of the thermally shocked specimens were identified by an x-ray diffractometer (XRD, X'pert PRO MPD, Philips, Netherlands) with Cu K_α radiation at 40 kV and 35 mA. The 2θ ranged from 5° to 80° with a scanning speed of 20° per minute.

3. Results

3.1 Residual Tensile Properties

The degradation of the tensile properties caused by the thermal shock cycles was determined by measuring the retained strength and modulus after thermally quenching the specimens from 900 to 300 °C for different cycles. Figure 3 shows two typical monotonic tensile stress-strain curves of 2D-C/SiC unshocked and thermally shocked for 60 cycles. It can be found that the strength and the modulus decreased significantly for the shocked composite. The degradation of the mechanical properties is closely related to the damage within the composite, which will be discussed later.

Figure 4 summarizes the residual tensile properties of the 2D-C/SiC composites experiencing different thermal shock cycles. The composite retains its strength within 20 thermal shock cycles. The strength decreases gradually when it is thermally shocked for more than 20 cycles, as shown in Fig. 4(a). However, the variation of the modulus decreases gradually with increasing number of thermal shock cycles, as shown in Fig. 4(b). The residual strength and modulus of 2D-C/SiC thermally shocked for 60 cycles are about 63 and 45%, respectively, of those for the unshocked material. The degradation of the modulus means that the damage appeared after first thermal shock and accumulated gradually in the subsequent cycles. The damage mechanisms will be revealed later. The different responses of the strength and modulus to the thermal shock suggest that the modulus should be more sensitive to the damage caused by the thermal shock and be more suitable to characterize the damage induced by the thermal shock (Ref 20).

3.2 Mass Variation

Figure 5 shows the mass variation of the material with increasing thermal shock cycles. It is found that significant mass loss takes place after 2D-C/SiC experiences 20 thermal

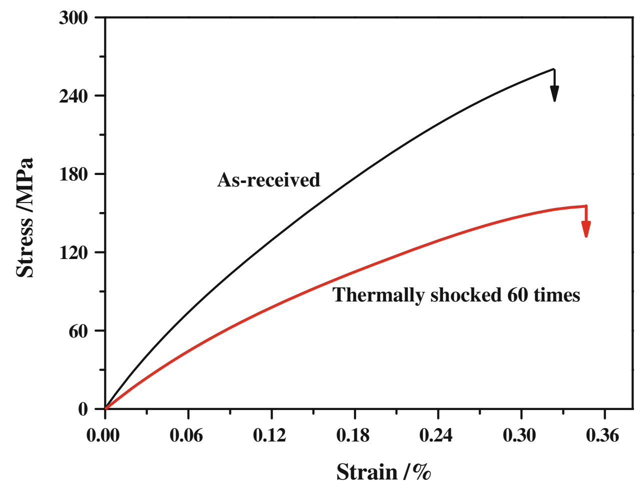

Fig. 3 Typical monotonic tensile stress-strain curves of the 2D-C/SiC unshocked and thermally shocked for 60 cycles

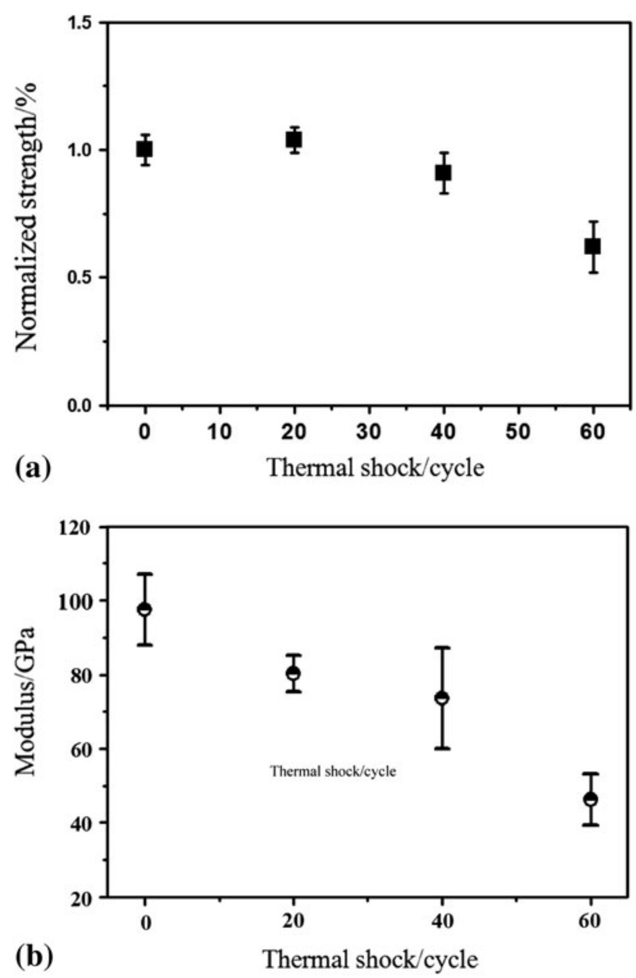

Fig. 4 Effects of the thermal shock on the tensile properties of 2D-C/SiC. (a) Strength and (b) modulus

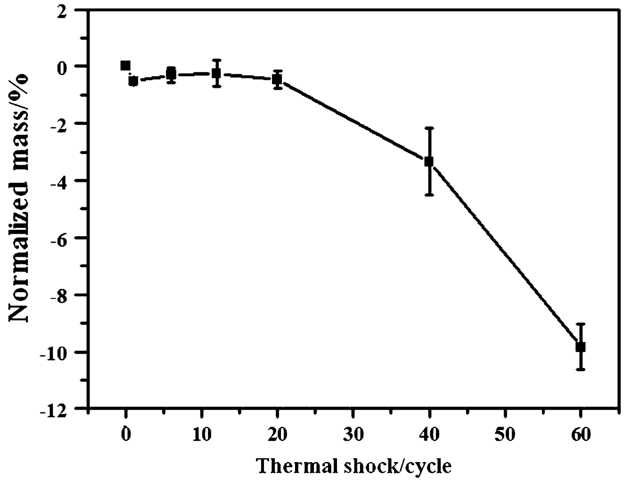

Fig. 5 Mass variations of 2D-C/SiC with increasing number of thermal shock cycles

shock cycles. The mass loss is about 9.8% after 60 thermal shock cycles. The oxidization of the carbon fibers or PyC interfaces should contribute to the mass loss in the composite. Comparing with the results in Fig. 4, it can be seen that the mass variations of 2D-C/SiC agree well with the variation of the residual strength.

3.3 Fracture Morphology

Figure 6 shows several typical tensile fracture morphologies of the thermally shocked 2D-C/SiC specimens. Extensive pullout of fibers can be found in the shocked 2D-C/SiC composite, regardless of cycles, as shown in Fig. 6(a), (b), and (d). Also, the debonding between the fiber bundles and the matrix can be seen on the fractured surfaces, suggesting a lower bonding strength of the fiber/matrix interface.

Most of the fibers in the near-surface zone can be seen in Fig. 6(a) for the specimen thermally shocked for 20 cycles. However, a large section of the fibers in the near-surface zone vanished for the specimen thermally shocked for 40 cycles, as indicated in Fig. 6(b). The reserved fibers became thinner and were characteristic of cone-like ends, indicated by a rectangle in Fig. 6(b) and displayed in a magnified image in Fig. 6(c). The cone-like end indicates that the carbon fibers are consumed by the manner of oxidation (Ref 21). With increasing number of cycles, the carbon fibers in the near-surface zone would be further oxidized, even completely vanished, as was evidenced in Fig. 6(d), in which the material experienced 60 thermal shock cycles. Meanwhile, it can be found from Fig. 6 that the fibers in the center of the fracture surface were not oxidized. Previous results show that the oxidization always occurs preferentially in the near-surface zone under a cracked surface (Ref 22). Therefore, the enhanced consumption of the carbon fibers in the near-surface zone suggest that more cracks and (or) larger cracks form when the composite was exposed to more number of thermal shock cycles. It is noticed that the amounts of the vanished carbon fibers in the near-surface zone well agree with the mass variation, shown in Fig. 5. A greater number of disappearing fibers correspond to larger mass loss. It can be primarily concluded that the oxidation of the fibers contributed to the mass loss in the composite.

3.4 Surface Morphology

Figure 7 shows the surface morphologies of the thermally shocked 2D-C/SiC. Dome-like columnar grains, a typical form of CVIed SiC, were found on the as-received material in Fig. 7(a). The characteristic of the dome-like morphology faded with the increasing number of thermal shock cycles. For example, the dimensions of the "domes," shown in Fig. 7(b) for the specimen thermally shocked for 20 cycles, became smaller than those on the unshocked specimen. After being thermally shocked for more than 20 cycles, the dome-like morphology completely disappeared, as seen in Fig. 7(c) and (d). In order to discover the mechanism in the morphological change, the composition of the specimen surface was examined by EDS. The composition of the surface is mainly composed of Si and O for the thermally shocked specimens. To further confirm the phase changes on the materials, the phases on the surface of the thermally shocked specimens were identified by XRD, as shown in Fig. 8. The intensity of SiC peaks becomes weaker for the specimens experiencing more number of thermal shock cycles. Meanwhile, the greater the amount of SiO_2 that appears, the stronger the intensities of its peaks become. This result indicates that a greater amount of SiC was oxidized and transformed to SiO_2 when the composite experienced more number of thermal shock cycles. Meanwhile, a small amount of B_2O_3 can also be identified. Therefore, the morphological change of the thermally shocked specimens, as shown in Fig. 7(b), was caused by the formation of the oxides (SiO_2 and B_2O_3). Apart from the oxidation of the coating, cracking and peeling off of the coating can also be found on the specimen thermally shocked for 40 cycles, as indicated by an arrow in Fig. 7(c). Oxygen in the air could diffuse into the composite through such damage and reacted with the carbon fibers. As a result, the carbon fibers were oxidized, as shown in Fig. 6(b) and (c). The oxidation and cracking of the coating would be more severe for the composite thermally shocked for more number of cycles, as can be evidenced in Fig. 6(d), where the carbon fibers in the near-surface zone were completely consumed. The consumption of the carbon fibers can also be deduced from the broken bubbles, on the specimen thermally shocked for 60 cycles, as seen in Fig. 7(d). Such morphology is produced by the escapement of a large amount of gas phases beneath the surface. According to above results, the resultant gas should be carbon dioxide (CO_2) or carbon monoxide (CO), produced by the reaction of carbon fiber with oxygen during thermal shock.

3.5 Microstructural Observations

As stated above, the degradation of the tensile properties and mass variation are closely related to the microstructural changes of the composite during the thermal shock. Figure 9 reveals the microstructures of 2D-C/SiC composites experiencing 20 thermal shock cycles. A layer of dark phase, shown in Fig. 9(a), was found on the surface of the specimen. According to the results in Fig. 8, the dark phase can be considered as some glassy oxides (SiO_2 and B_2O_3). With increasing number of thermal shock cycles, more coating was oxidized and more oxides were generated. Furthermore, the oxidizing gas could penetrate into the specimens and react with carbon fibers through the damaged coating. Accordingly, non-uniform oxidized carbon fibers and consumption of fibers can be seen for the specimen thermally shocked for 60 cycles. These

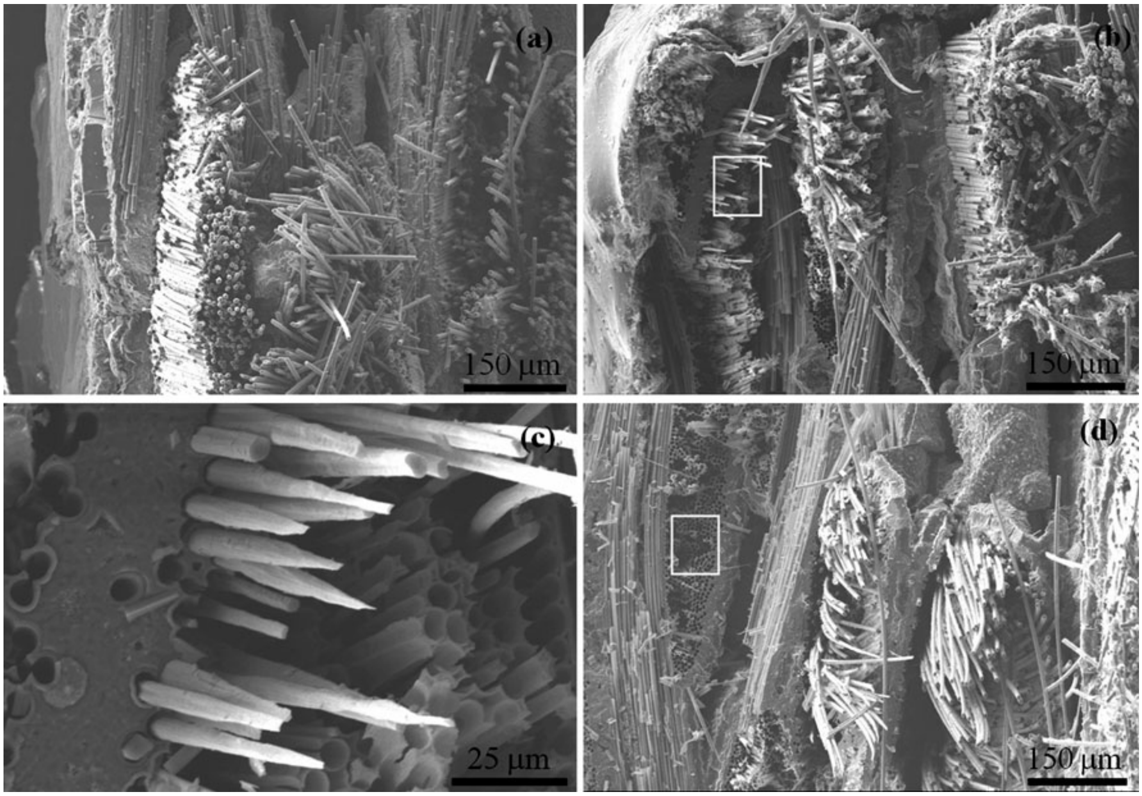

Fig. 6 Fractured morphologies of the thermally shocked 2D-C/SiC. (a) 20 cycles, (b) 40 cycles, (c) A magnified picture of the area indicated in (b), and (d) 60 cycles

observations well agree with the results on fractured surfaces as shown in Fig. 6.

4. Discussion

It can be found from Fig. 3 and 4 that the thermal shock had great effects on the degradation in mechanical properties of the 2D-C/SiC. The degradation of the mechanical properties is closely related to the evolution of the damage in the composite (Ref 20, 23). In the thermal shock test, the surface of the composite was cooled to the lower temperature rapidly ($\sim$15 s in the present investigation), whereas the interior of the composite remained at higher temperature. The temperature gradient created a tensile stress at the surface and a compressive stress in the interior. The stress gradient could result in cracking and peeling off of the coating, as seen in Fig. 7(c). Moreover, SiC matrix has a greater thermal expansion coefficient (CTE) (4.6×10^{-6}/K) than the longitudinal CTE of the carbon fiber (0.5×10^{-6}/K). The sudden change of temperature in the materials also produced tensile stress in the matrix. With increasing number of the thermal shock cycles, the tensile stress in the matrix was accumulated and could reach the critical value sufficient to create matrix cracks in the matrix (Ref 24-26). Meanwhile, the number and dimension of the matrix cracks increased as the number of thermal cycles was increased (Ref 14). In addition, the radial CTE of the fiber (7.0×10^{-6}/K) is greater than that of the matrix. Tensile stress was produced on the fiber/matrix interface during the successive quenching operations from 900 to 300 °C. The repeated tensile stresses in the interfaces of fiber (bundles)/matrix could weaken the interface bonding, and lead to delamination and debonding of fiber/matrix interface. Moreover, the weakened interface strength between the fiber and matrix could promote longer pullout length of the fibers (Ref 27). The extensive pullout of carbon fibers in Fig. 6 confirms that the interface bonding between the fibers (or bundles) and the matrix was degraded by the thermal shock.

Since the composite was damaged more severely when exposed to more number of cycles, the mechanical properties of the composite degraded with increasing number of thermal shock cycles, as shown in Fig. 3 and 4. As for the different sensitivities to the damage for the strength and modulus (Fig. 4), it can also be understood from the evolution of damage (Ref 28, 29). The early thermal shock damage is mainly in the form of matrix microcracks, located at the surface region of the composite. The subtle development of these microcracks was undetectable by the conventional strength tests, such as flexural strength (Ref 20) and tensile strength in the present study, whereas, it could be sensitively detected by modulus (Ref 28, 29). Apart from the damage mentioned above, the oxidation of the coating and the matrix also took place, and it enhanced the damage in the 2D-C/SiC material. SiC and B_4C can be oxidized at about 600 and 400 °C, respectively. The temperature difference investigated in the present study was 300-900 °C. Hence, SiC and B_4C might react

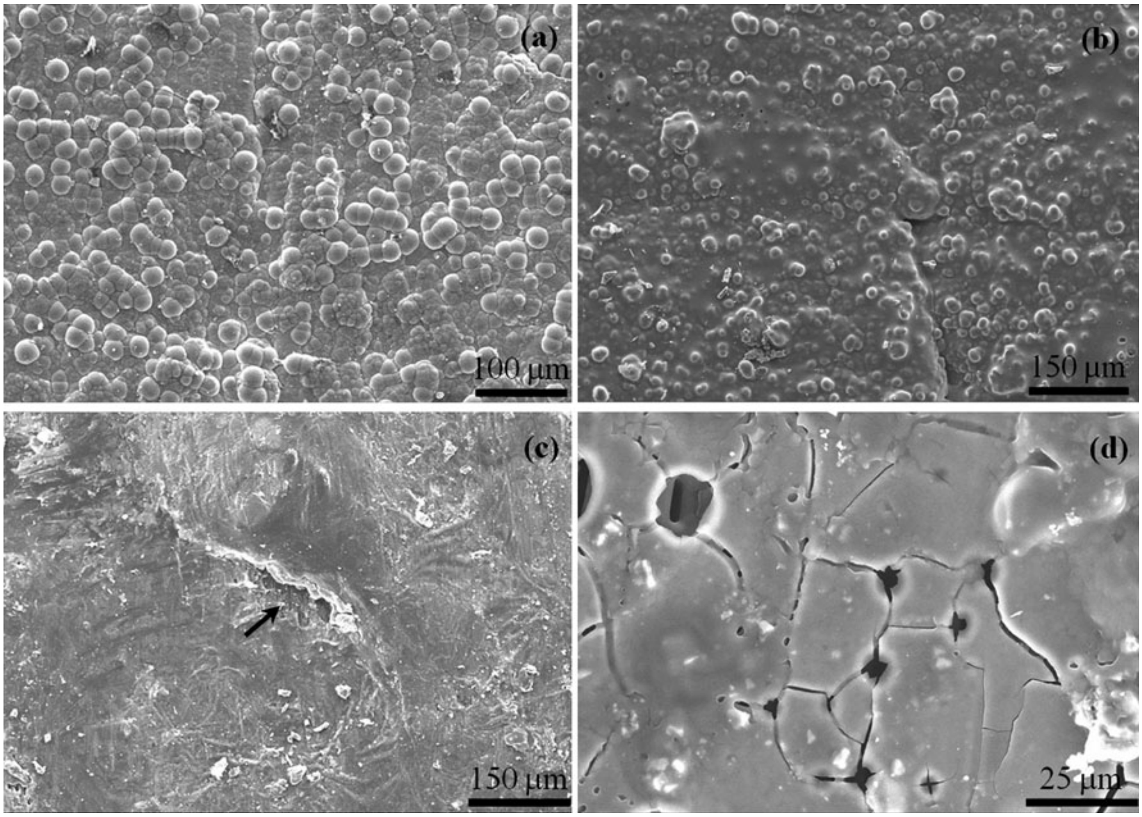

Fig. 7 Surface morphologies on the 2D-C/SiC thermally shocked for different cycles. (a) Dome-like morphology on the as-received specimen. (b) Surface morphology of the specimen experiencing thermal shocks for 20 cycles. (c) Peeling off of the coating of the specimen thermally shocked for 40 cycles, indicated by an arrow. Meanwhile, the dome-like pattern disappeared. (d) Surface morphology on the specimen thermally shocked for 60 cycles

with oxygen in air at 900 °C. The possible reactions are listed as follows:

$$SiC(s) + 2O_2(g) = SiO_2(l) + CO_2(g) \quad \text{(Eq 1)}$$

$$B_4C(s) + 4O_2(g) = 2B_2O_3(l) + CO_2(g) \quad \text{(Eq 2)}$$

According to the thermochemical data (Ref 30), the Gibbs free energies of reaction (1) and (2) are about −1028 and −2298 kJ/mol, respectively. The negative values of the Gibbs free energy indicate that the above oxidation reactions could take place spontaneously when the specimens were heated to 900 °C during the thermal shock. If the coating was intact, then only superficial oxidation occurred and the oxygen could not access the carbon fiber or the interface. In other words, the superficial oxidation had negligible effects on the degradation of the tensile properties. Therefore, the composite almost retained the original strength within the 20 cycles because the oxidizing atmosphere could not attack the fibers and interfaces directly (Ref 31). Once the composite was attacked by the greater number of thermal shock cycles, the heavily cracked coating and matrix no longer protected the carbon fibers and PyC interface from oxidization. The matrix cracks in the coating and matrix offered the channels for the oxidizing atmosphere to diffuse into the composite. The carbon fibers can be oxidized at temperatures as low as 400 °C. The reaction between of the carbon fibers and oxygen could produce CO_2 or CO. The products would escape from the composite in the form of gases. More severe cracking and oxidation could be

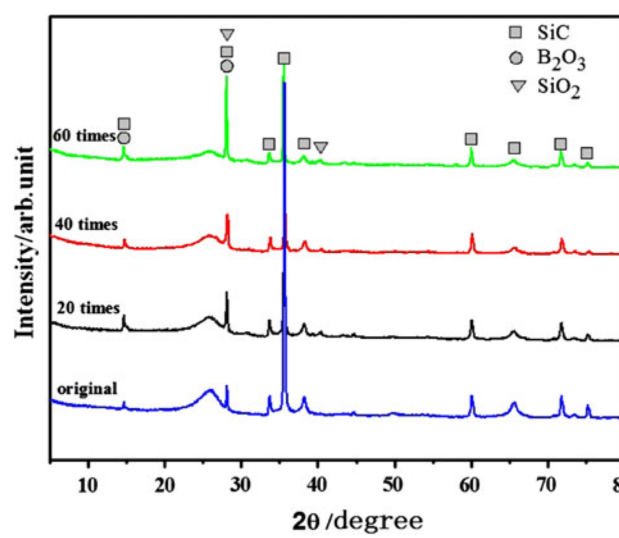

Fig. 8 XRD patterns of the 2D-C/SiC experiencing different thermal shock cycles

anticipated for the composite thermally shocked for greater number of cycles. As a result, a greater number of carbon fibers were found to be consumed, as shown in Fig. 6. The removal of the carbon fibers and interfaces in the near-surface zone, as shown in Fig. 6 and 9, also confirms the discussion. The oxidation of the PyC interphase reduced the interfacial shear

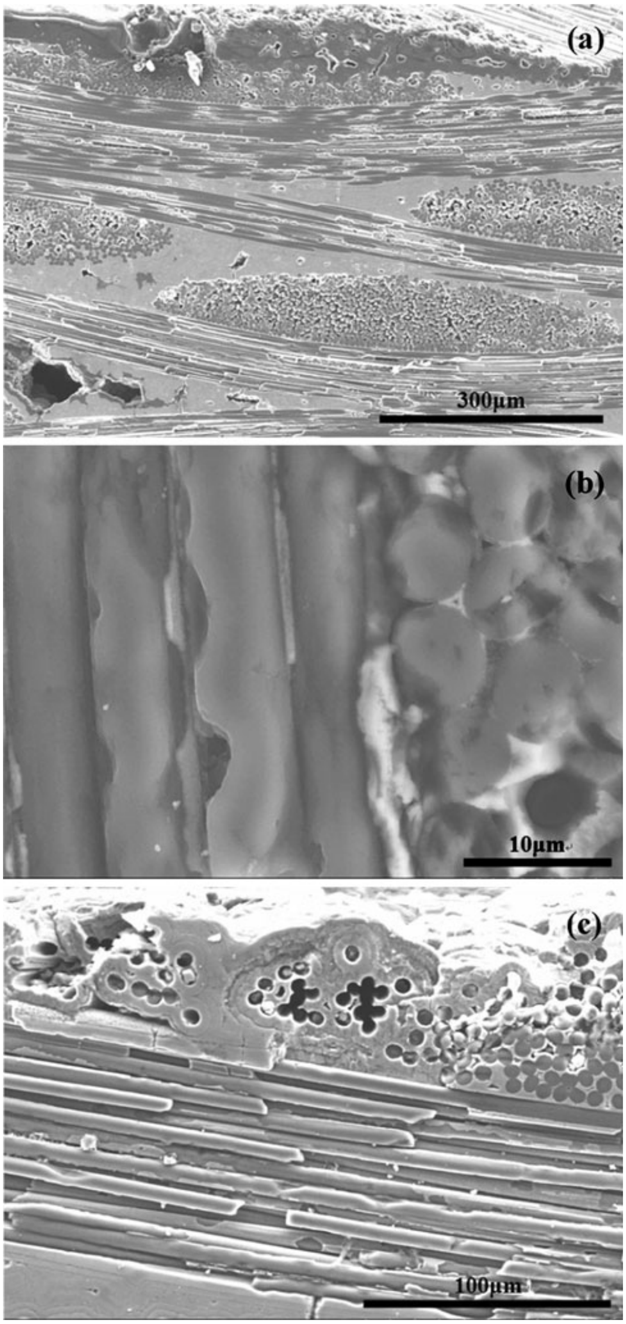

Fig. 9 Microstructure of the polished cross section of the thermally shocked 2D-C/SiC. (a) Superficial oxidization for the specimen, 20 cycles; (b) Oxidation on the carbon fibers, 60 cycles; and (c) Burned-out carbon fibers in the near-surface zone, 60 cycles

strength, and decreased the matrix cracking stress (Ref 6). Accordingly, the cracks in the matrix could be induced at lower stress. Furthermore, the consumption of the fiber could decrease the effective loaded area of the fibers. The damage caused by the oxidization also contributes to the degradation of the strength.

In addition, the oxidation is responsible for the mass variation of 2D-C/SiC, as shown in Fig. 5. The production of CO_2 (or CO) leads to the weight loss; however, the reactions of SiC and B_4C with oxygen result in the mass gain. Therefore, the mass loss is negligible as long as the carbon fibers are protected by the coating and matrix, (Ref 18). According to the results shown in Fig. 6 and 7, the carbon fibers are almost intact within 20 cycles, although the coating was oxidized. Therefore, no significant mass loss, as shown in Fig. 5, can be found for the composite thermally shocked until 20 cycles. The coating and matrix with accumulated cracks could not prevent the oxygen from penetrating into the composite with the increasing number of thermal shock cycles. As a result, the carbon fibers in the near-surface zone were gradually oxidized when the composite was thermally shocked for more than 20 cycles. The gaseous product could dominate in the contribution to the mass loss of the composite exposed to more than 20 cycles (Fig. 5). Accompanied by the escapement of the gases, broken bubbles or pores were left on the surface, as shown in Fig. 7(d). Similar morphology was also to be found on SiC/SiC (Ref 32). Therefore, measures of protecting the carbon fibers from oxidation are of critical importance to improve the thermal shock resistance of the composite.

5. Conclusion

(1) 2D-C/SiC retains its tensile strength when thermally shocked until 20 cycles. However, the residual modulus degrades gradually with the increasing number of thermal shock cycles. The degradation in the mechanical properties is closely related to the damage produced by the thermal shock.

(2) The bonding strengths of fiber/matrix or coating/composite interfaces are weakened by the accumulated thermal stress during the thermal shock, as can be verified by the extensive pullout of fibers, the delamination between the fiber bundles and matrix, and the peeling off of the coating.

(3) The matrix cracking and delamination of the coating/composite or fiber/matrix interface offer the channels where oxygen can penetrate into the composite and consumes the carbon fibers gradually with the increasing number of thermal shock cycles. The prevention of reaction of carbon fiber with oxygen plays an important role in enhancing the resistance to the thermal shock of the composite.

Acknowledgments

The authors would like to sincerely appreciate the reviews for the valuable comments and kind suggestions. This study is financially supported by the National Science Foundation of China (51172182), the Program for New Century Excellent Talents in University (NCET-08-0460), the Basic Research Foundation of NWPU (JC20100227), and the "111" Project (B08040).

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